The empirical recurrence for OEIS A320369, recovered from its tail

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Abstract

OEIS A320369 records "Empirical recurrence of order 70" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 227$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{70} = -37748736 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A320369 is "Number of nX6 0..1 arrays with every element unequal to 0, 1, 2, 3 or 4 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$32, 2048, 80568, 3027408, 114844160, 4358279956, 165355751224, 6274017360108, \ \ldots$$

The entry states:

> Empirical recurrence of order 70 (see link above)

As of the "Last modified" line on the live entry (Oct 11 2018 14:11:14, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 227$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 227$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 454$ exact terms from $n = 4$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{70} c_i\, a(n-i),$$

c_1	64	c_{19}	60135276732826	c_{37}	8117464164660499150	c_{55}	−257080102874496
c_2	−1218	c_{20}	−233607374321848	c_{38}	9108301200643440547	c_{56}	−1928761841053952
c_3	9486	c_{21}	24694891362623	c_{39}	6861244314314327270	c_{57}	−1364420108291328
c_4	−29455	c_{22}	604473941255429	c_{40}	3549242264565319147	c_{58}	−1360366973912832
c_5	−43644	c_{23}	−185523715341832	c_{41}	−255821751769197465	c_{59}	−122018813452288
c_6	1427115	c_{24}	−2341631978554731	c_{42}	−3085660204326715680	c_{60}	19146236307456
c_7	−10158332	c_{25}	−479313836290466	c_{43}	−3932838107866718541	c_{61}	26768897632256
c_8	19579475	c_{26}	−14632511495320388	c_{44}	−3639559622884033651	c_{62}	26225763647488
c_9	62150137	c_{27}	33094058073253373	c_{45}	−1749510941201671639	c_{63}	9944516849664
c_{10}	−708020437	c_{28}	5130672372432238	c_{46}	−1070924056004130592	c_{64}	10170750627840
c_{11}	3904964497	c_{29}	130625415334151420	c_{47}	64018221646973760	c_{65}	28323381248
c_{12}	−8984028689	c_{30}	−301371638570268177	c_{48}	−77598972017891112	c_{66}	−956514648064
c_{13}	10372647062	c_{31}	607989390324667430	c_{49}	196129317852875728	c_{67}	49735139328
c_{14}	101689936262	c_{32}	265860692072054666	c_{50}	93149918492991264	c_{68}	10363338752
c_{15}	−537935563807	c_{33}	2190912332287174210	c_{51}	90562305902801088	c_{69}	−165675008
c_{16}	2656112935900	c_{34}	3239351105695443961	c_{52}	44834681480241520	c_{70}	−37748736
c_{17}	−1249087570063	c_{35}	4360977343674148531	c_{53}	22896485924452288		
c_{18}	−10220782830894	c_{36}	7698500926699338163	c_{54}	−321304533342784		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A320369 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{70-1} - \cdots - c_{70}$. Its constant term is $-c_{70} = 37748736$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with

$\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -37748736.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A320369.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.